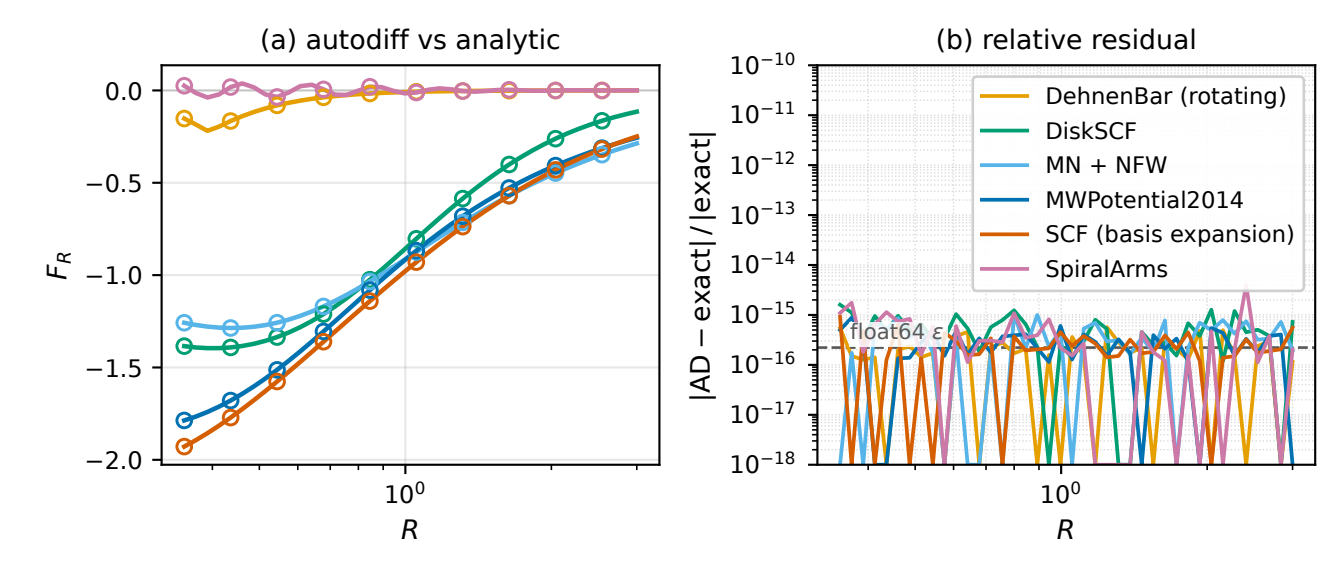
(a) autodiff vs analytic
F_R
0.0
−0.5
−1.0
−1.5
−2.0
10^0
R
(b) relative residual
$|\mathrm{AD} - \mathrm{exact}| / |\mathrm{exact}|$
10^{-10}
10^{-11}
10^{-12}
10^{-13}
10^{-14}
10^{-15}
10^{-16}
10^{-17}
10^{-18}
float64 ε
DehnenBar (rotating)
DiskSCF
MN + NFW
MWPotential2014
SCF (basis expansion)
SpiralArms
10^0
R